ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Algebra Toolkit

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

ALGEBRA TOOLKIT · CH2 · L1

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The Map

This lesson collapses Algebra Toolkit into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Name the target

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Write the first line

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Calculate carefully

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Answer in the requested form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use index laws

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Factorise expression

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 106 website questions, 106 checked solutions, 3 local PDFs read.

01 Name the target

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebra Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3(2x-5) = \frac{9-x}{2}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q7

(a)

This is a reasoning step: write one clear sentence, then continue the calculation.

(b)

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: (a) $m^2 - 3m - 40$, (b) $5y(1+4y)$, (c) 1, (d) $x = 3$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

> 7 (a) Expand and simplify (m - 8)(m + 5) (2) (b) Factorise fully 5y + 20y2 (2) (c) Simplify (p2 + 3)0 (1) (d) Solve 3(2x - 5) = 9 2 -x Show clear algebraic working. x = (4)

YOUR SOLUTION

02 Write the first line

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebra Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{x^9}{x^2} = x^{9-2} = x^7$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q1

(a)

$$\frac{x^9}{x^2} = x^{9-2} = x^7$$

(b)

$$\frac{7^8 \times 7^4}{7^3} = 7^{8+4-3} = 7^9$$

Finish: (a) x^7, (b) 7^9.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

1 (a) Simplify x x 9 2 (1) (b) Write 7 7 7 8 4 3 times as a single power of 7 (2)

YOUR SOLUTION

03 Calculate carefully

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebra Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$2^7 \times 4^5 = 2^7 \times (2^2)^5 = 2^{17}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2023 P2H Q12

(a)

$$2^7 \times 4^5 = 2^7 \times (2^2)^5 = 2^{17}$$

Use index laws

$$4^x = (2^2)^x = 2^{2x}$$

Finish: (a) $x = \frac{17}{2}$, (b) $25p^4y^{16}$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

12 2 4 4 7 5 x (a) Calculate the value of x x = (2) (b) Simplify fully 125 6 24 2 3 p y (2)

YOUR SOLUTION

04 Answer in the requested form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebra Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{4}{x-2}-\frac{3}{x+1}=\frac{4(x+1)-3(x-2)}{(x-2)(x+1)}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q12

(a)

$$\frac{4}{x-2}-\frac{3}{x+1}=\frac{4(x+1)-3(x-2)}{(x-2)(x+1)}$$

(b)

$$2x(x-5)(x-3)=2x(x^2-8x+15)$$

Finish: (a) $\frac{x+10}{(x-2)(x+1)}$, (b) $2x^3-16x^2+30x$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

> 12 (a) Express 4 2 3 1 x x - - + as a single fraction. Give your answer in its simplest form. (3) (b) Expand and simplify 2x(x - 5)(x - 3) (3)

YOUR SOLUTION

05 Use index laws

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebra Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$10x^2 + 23x + 12 = (5x + 4)(2x + 3)$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q21

Factorise expression

$$10x^2 + 23x + 12 = (5x + 4)(2x + 3)$$

Use index laws

$$\frac{10x^2 + 23x + 12}{4x^2 - 9} = \frac{(5x + 4)(2x + 3)}{(2x - 3)(2x + 3)}$$

Finish: (a) $\frac{5x+4}{2x-3}$, (b) $n = 5y - 1$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

> 21 (a) Simplify fully 10 23 12 4 9 2 2 x x x + + - (3) 2 2 8 4 2 3 2 5 y y y n times = + (b) Find an expression for n in terms of y. Show clear algebraic working and simplify your expression. (4)

YOUR SOLUTION

06 Factorise expression

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Algebra Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$10x^2 + 23x + 12 = (5x + 4)(2x + 3)$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q21

Factorise expression

$$10x^2 + 23x + 12 = (5x + 4)(2x + 3)$$

Use index laws

$$\frac{10x^2 + 23x + 12}{4x^2 - 9} = \frac{(5x + 4)(2x + 3)}{(2x - 3)(2x + 3)}$$

Finish: (a) $\frac{5x+4}{2x-3}$, (b) $n = 5y - 1$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

> 21 (a) Simplify fully 10 23 12 4 9 2 2 x x x + + - (3) 2 2 8 4 2 3 2 5 y y y n times = + (b) Find an expression for n in terms of y. Show clear algebraic working and simplify your expression. (4)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Name the target
"work out", "show", "hence", a familiar exam phrase	Write the first line
"work out", "show", "hence", a familiar exam phrase	Calculate carefully
"work out", "show", "hence", a familiar exam phrase	Answer in the requested form
"work out", "show", "hence", a familiar exam phrase	Use index laws
"work out", "show", "hence", a familiar exam phrase	Factorise expression

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.